

RESUME

Name- Komal Barhate

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OBJECTIVE

Engineering student focused on building real-world, scalable applications using full-stack development and AI. Passionate about solving practical problems and deploying functional systems that deliver measurable impact.

EDUCATION

Bachelor of Engineering (Pursuing) — Computer Engineering (Data Science) *Expected 2028*
Vidyavardhini's College of Engineering and Technology, Vasai West

TECHNICAL SKILLS

Languages:	Python, HTML, JavaScript
Frameworks & Tools:	Flask, Streamlit, FastAPI, React, Git, GitHub, VS Code
Technologies:	Full-Stack Development, AI/ML, Image Forensics, Network Analysis, REST APIs, SQLite
Libraries:	NumPy, pandas, scikit-learn, OpenCV, Pillow, Scapy, Socket.IO

PROJECTS

MedEase – Hospital OPD Management System *medease-923t.onrender.com*

- Built a full-stack OPD system using Flask and SQLite for end-to-end hospital workflow management
- Patient appointment booking with live queue tracking and real-time status updates
- Integrated UPI QR-based payment system for seamless in-app transactions
- Admin dashboard for monitoring appointments, operations, and analytics
- Implemented AI chatbot (MedBot) for patient assistance and symptom triage

Digital Image Tampering Detection *github.com/komallbarhate/tamper-detect*

- Detects suspicious edits in digital photos using image forensics — scores images (0–1) and highlights regions with inconsistent JPEG compression, common when areas are pasted, retouched, or re-saved
- Generates an ELA-based heatmap overlay and a scale-gap map (two ELA passes at different JPEG strengths) to pinpoint recompression anomalies
- Produces an auto-generated forensic brief with plain-language bullets, benign explanations, and a shareable case signature
- Supports per-run SQLite scan history, CLI-based analysis, and an optional FastAPI endpoint for external integrations
- Deployed live on Streamlit Community Cloud; tech: Python, NumPy, Pillow, OpenCV, FastAPI, SQLite, Streamlit

NetMon – Real-Time Network Monitoring System *github.com/komallbarhate/network-monitoring*

- Built a real-time network monitoring tool using Python and Flask that captures live traffic, analyzes packets for anomalies, and displays metrics on a dynamic web dashboard
- Integrated ML-based anomaly detection (scikit-learn) alongside rule-based logic to flag suspicious network activities
- Live dashboard powered by Socket.IO for real-time updates with a professional light-themed interface and micro-animations
- Role-based access control with Admin and Viewer roles using Flask-Login and Flask-SQLAlchemy
- Tech stack: Python, Flask, Flask-SocketIO, Scapy, pandas, NumPy, scikit-learn, HTML/CSS/JS

Sci-Fi Vitals Anomaly Detection *github.com/komallbarhate/scifi-vitals-anomaly-detection*

- ML pipeline for ICU vital-sign anomaly detection built for a hackathon challenge
- Reads vital-sign time series, engineers anomaly-focused features, runs a detection model, and produces a submission file (case_id, time_sec, anomaly_score)

- Modular architecture: separate modules for feature engineering, modeling, pipeline orchestration, and end-to-end smoke testing
- Tech stack: Python, pandas, NumPy, scikit-learn

Inkwell – Full-Stack Blogging Platform

inkwelllll.loveable.app

- Modern, full-stack blog platform where writers can publish posts and readers can discover stories; built with a focus on clean typography and a distraction-free editorial experience
- Features include a post editor, public feed, user profiles with customizable bio, personal dashboard for post management, and Google + email/password authentication
- Secured with row-level security — users can only modify their own data while published posts remain publicly accessible
- Tech stack: React 19, TanStack Start (SSR), Vite 7, Tailwind CSS v4, shadcn/ui, Loveable Cloud (managed Postgres + Auth), Google OAuth

EXPERIENCE

Sub Admin Head – Literati Club

VCET, Vasai West

- Managed documentation, official reports, and event registrations
- Drafted official communications and administrative documents for the club
- Coordinated event workflows and operations across teams

INTERESTS

Artificial Intelligence | Product Development | Scalable Web Applications | UI/UX Design | Cybersecurity